ABDELRAHMAN ALAA AZAB

Computer & Intelligent Systems Engineer

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SUMMARY

Computer and Intelligent Systems Engineering graduate with experience in full-stack web development. Experienced in designing and developing scalable RESTful APIs using Node.js and Express.js, implementing secure authentication and authorization systems, and optimizing database performance. Proven ability to work with data analysis, machine learning, and computer vision, including data preprocessing, model training, and model evaluation using Python and Scikit-learn. Hands-on experience in building image classification systems using deep learning techniques. Solid understanding of cybersecurity fundamentals, secure software development practices, and modern web security principles. Seeking to contribute to real-world, production-level software systems.

EDUCATION

Bachelor of Engineering in Computer & Intelligent Systems Engineering

Future University in Egypt (FUE), Faculty of Engineering and Technology

Graduated Fall 2025

CERTIFICATIONS & PROFESSIONAL TRAINING

Full-Stack Web Development Bootcamp — Udemy/SEF Academy (HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, REST APIs, Authentication, MVC)

Data Science & Analytics — HP Foundation (Data analytics workflows and data-driven decision-making)

Machine Learning — AITB (Supervised & unsupervised learning, feature engineering, model evaluation (Python, Scikit-learn))

Flutter Development — Udemy (Cross-platform mobile apps using Flutter, Dart, REST APIs, and Firebase)

LANGUAGE

ARABIC — NATIVE

ENGLISH — FLUENT

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, Dart, C++
Frameworks & Libraries: Node.js, Express.js, React.js, Flutter
Web Development: HTML5, CSS3, RESTful APIs, MVC Architecture, Authentication (JWT & Sessions), Middleware, API Security
Databases: MongoDB, MySQL, Firebase Firestore
AI & Machine Learning: Scikit-learn, Pandas, NumPy, Data Preprocessing, Model Training & Evaluation
Computer Vision: Convolutional Neural Networks (CNNs), Image Processing, Data Augmentation
Cybersecurity: Cryptography Fundamentals, DES Algorithm, Secure Authentication, Web Security Principles (SQL Injection, XSS)
Tools & Platforms: GitHub, Postman, VS Code, Jupyter Notebook, Firebase
Core Concepts: Full-Stack Development, OOP, Data Structures & Algorithms, Software Engineering Principles, Agile Methodologies, Database Design, Computer Networks Fundamentals

PROFESSIONAL EXPERIENCE

AI-Based Interactive Virtual Clinic - Graduation Project – Software Developer

- Designed and developed a scalable AI-powered virtual clinic platform enabling users to search for medical services and book appointments efficiently.
- Developed backend services for appointment scheduling, user authentication, and medical service management.
- Implemented real-time communication features to support seamless interaction between patients and healthcare providers.
- Built an AI-driven chatbot using Natural Language Processing (NLP) techniques to handle inquiries related to diseases, symptoms, and medical tests.
- Designed and integrated advanced search and filtering algorithms to enhance accessibility and overall user experience.
- Ensured system scalability, data accuracy, and compliance with healthcare data handling standards through structured architecture and Agile collaboration.

Additional Academic Projects

- **Traffic Sign Recognition System:** Built and trained a CNN-based model for multi-class traffic sign recognition using image preprocessing and data augmentation.
- **Cryptography Algorithms Project:** Implemented classical ciphers and the DES algorithm using key generation and substitution–permutation networks.
- **Robot Arm Control System:** Built an Arduino-based robot arm supporting forward, reverse, and directional movement using motor drivers and programmed control logic.
- **Autonomous Line-Following Vehicle:** Designed and implemented a sensor-based autonomous car using IR sensors and Arduino to follow a predefined track.
- **Fire Alarm System:** Developed an embedded fire detection system using Arduino and sensors with real-time alert functionality.